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Error Analysis

Carrying out error analysis

Look at dev examples to evaluate ideas



90% accuracy
→ 10% error

Should you try to make your cat classifier do better on dogs? ←

Error analysis:

- Get ~100 mislabeled dev set examples. → 5-10 min
- Count up how many are dogs.

→ 5%
5/100

10%
↓
9.5%

"ceiling"

→ 50%
50/100

10%
↓
5%

Evaluate multiple ideas in parallel

Ideas for cat detection:

- Fix pictures of dogs being recognized as cats ←
- Fix great cats (lions, panthers, etc..) being misrecognized ←
- Improve performance on blurry images ←

Image	Dog	Great Cats	Blurry	Instagram	Comments
1	✓			✓	Pitbull
2			✓	✓	
3		✓	✓		Rainy day at zoo
⋮	⋮	⋮	⋮	⋮	
% of total	<u>8%</u>	<u>43%</u>	<u>61%</u>	<u>12%</u>	



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Error Analysis

Cleaning up
Incorrectly labeled
data

Incorrectly labeled examples

x							
y	<u>1</u>	<u>0</u>	<u>1</u>	<u>1</u>	<u>0</u>	<u>1</u>	1

Training set.

↑

DL algorithms are quite robust to random errors in the training set.

Systematic errors

Error analysis

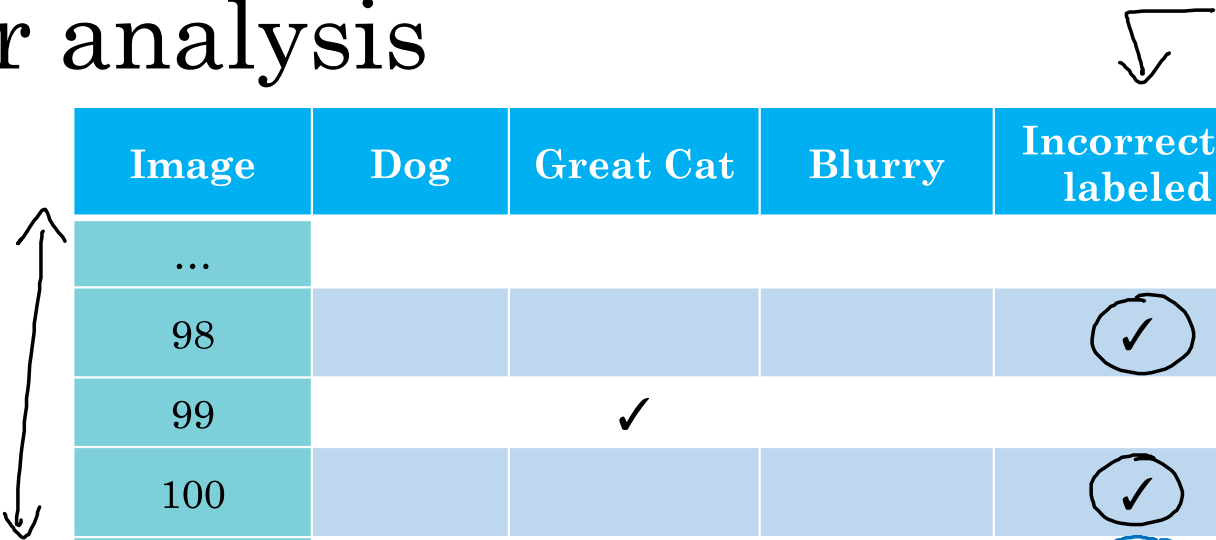


Image	Dog	Great Cat	Blurry	Incorrectly labeled	Comments
...					
98				✓	Labeler missed cat in background
99		✓			
100				✓	Drawing of a cat; Not a real cat.
% of total	<u>8%</u>	<u>43%</u>	<u>61%</u>	<u>6%</u>	

Overall dev set error 100%

Errors due incorrect labels 0.6% ←

Errors due to other causes 9.4% ←

↑

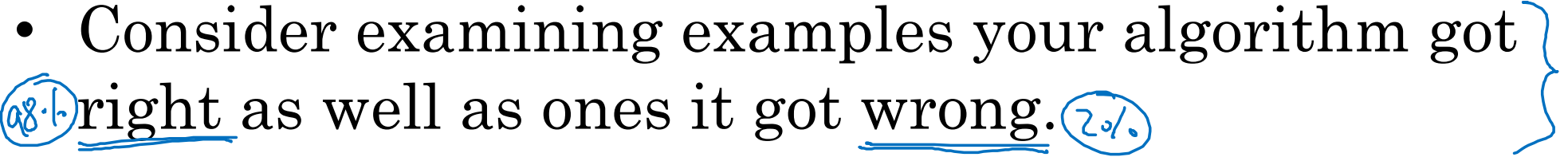
✓
2.0%
 ✓
0.6%
 1.4%
2.1%

1.9%

Goal of dev set is to help you select between two classifiers A & B.

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Correcting incorrect dev/test set examples

- Apply same process to your dev and test sets to make sure they continue to come from the same distribution
- Consider examining examples your algorithm got right as well as ones it got wrong.

- Train and dev/test data may now come from slightly different distributions.



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Error Analysis

Build your first system
quickly, then iterate

Speech recognition example



- • Noisy background
 - • Café noise
 - • Car noise

- • Accent
- • Far from
- • Young
- • Stutter
- • ...

Guideline:

**Build your first
system quickly,
then iterate**

- • Set up dev/test set and metric
- Build initial system quickly
- Use Bias/Variance analysis & Error analysis to prioritize next steps.



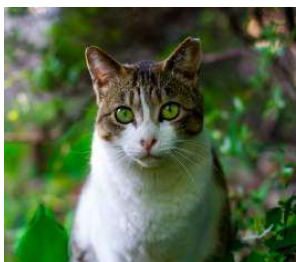
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Mismatched training
and dev/test data

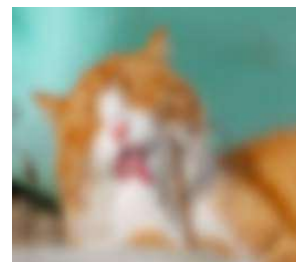
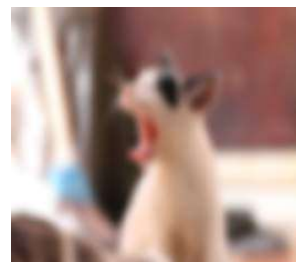
Training and testing
on different
distributions

Cat app example

Data from webpages



care about this
Data from mobile app



→ ≈ 200,000

210,000

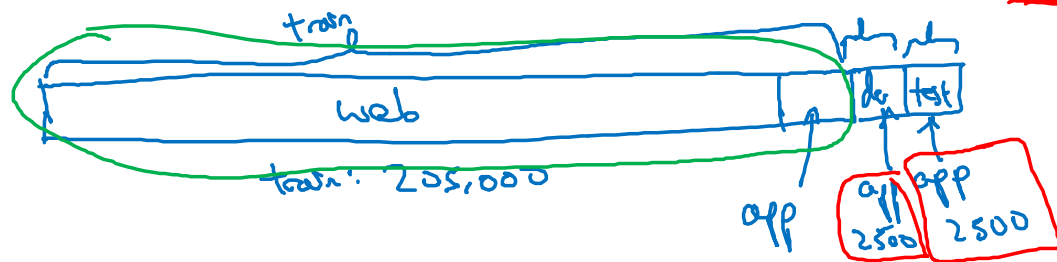
↓ shuffle

→ ≈ 10,000

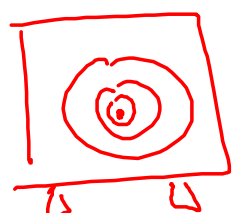
~~Option 1:~~



Option 2:



$$\frac{200K}{210K}$$



2381 - web
119 - mobile app



Speech recognition example

Speech activated rearview mirror



Training

Purchased data

$\downarrow \downarrow$
 x, y

Smart speaker control

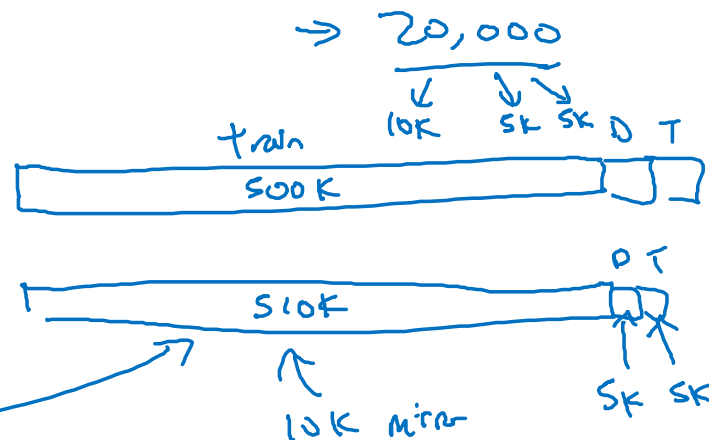
Voice keyboard

...

500,000 utterances

Dev/test

Speech activated
rearview mirror





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Mismatched training
and dev/test data

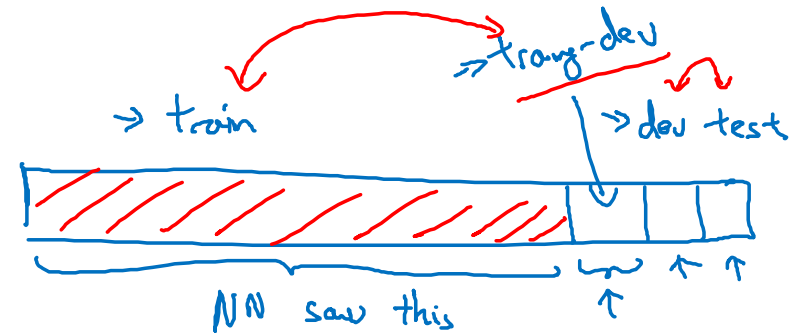
**Bias and Variance with
mismatched data
distributions**

Cat classifier example

Assume humans get $\approx 0\%$ error.

Training error 1%
 Dev error 10% $\downarrow 9\%$

Training-dev set: Same distribution as training set, but not used for training



Training error	1%		1%
→ Training-dev error	9%	↑ variance	1.5%
→ Dev error	10%		10% ↓ data mismatch
		Variance	
Human error - - -	0%	↑ Avoidable bias	10% ↑ Avoidable bias
Training error	10%	↓ bias	10% ↓ variance
Training-dev error	11%		11% ↑ Data mismatch
Dev error	12%		20%
	Bias		Bias + Data mismatch

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Bias/variance on mismatched training and dev/test sets

Human level	4%		4%
<u>Training</u> set error	7%	↑ avoidable bias	7%
<u>Training</u> - dev set error	10%	↑ variance	10%
→ Dev error	12%	↑ data mismatch	6%
→ Test error	12%	↑ degree of overfitting to dev set.	6%

More general formulation

Rearview mirror

	General speech recognition	Rearview mirror speech data.
Human level	"Human level" <u>4%</u>	<u>6%</u>
Error on examples trained on	 "Training error" 7%	 6%
Error on examples <u>not</u> trained on	 "Training-dev error" 10%	 "Dev/Test error" <u>6%</u>

↔ data mismatch ↔

↑ avoidable bias
↑ Variance



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Mismatched training
and dev/test data

Addressing data
mismatch

Addressing data mismatch

- • Carry out manual error analysis to try to understand difference between training and dev/test sets

E.g. noisy - car noise

street numbers

- • Make training data more similar; or collect more data similar to dev/test sets

E.g. Simulate noisy in-car data

Artificial data synthesis



+



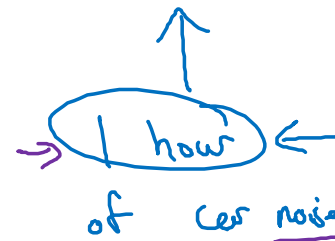
=



“The quick brown
fox jumps
over the lazy dog.”

↑
10,000 hours

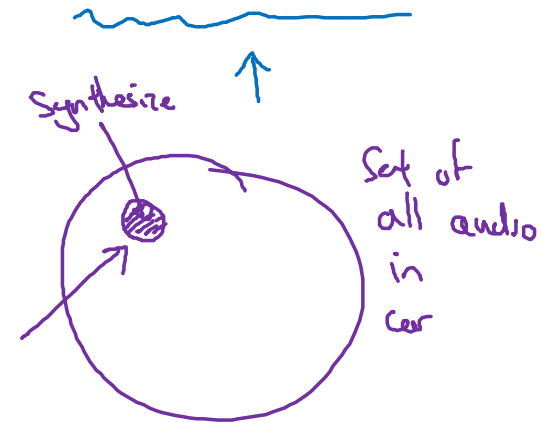
Car noise



Overfit to 1 hour of
car noise

↑
10,000 hours

Synthesized
in-car audio



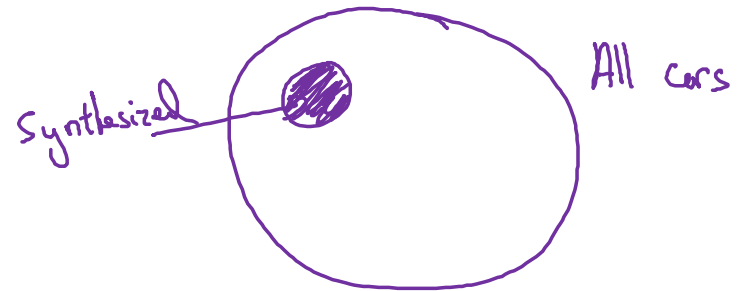
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Artificial data synthesis

Car recognition:



$\approx \underline{20}$ cars



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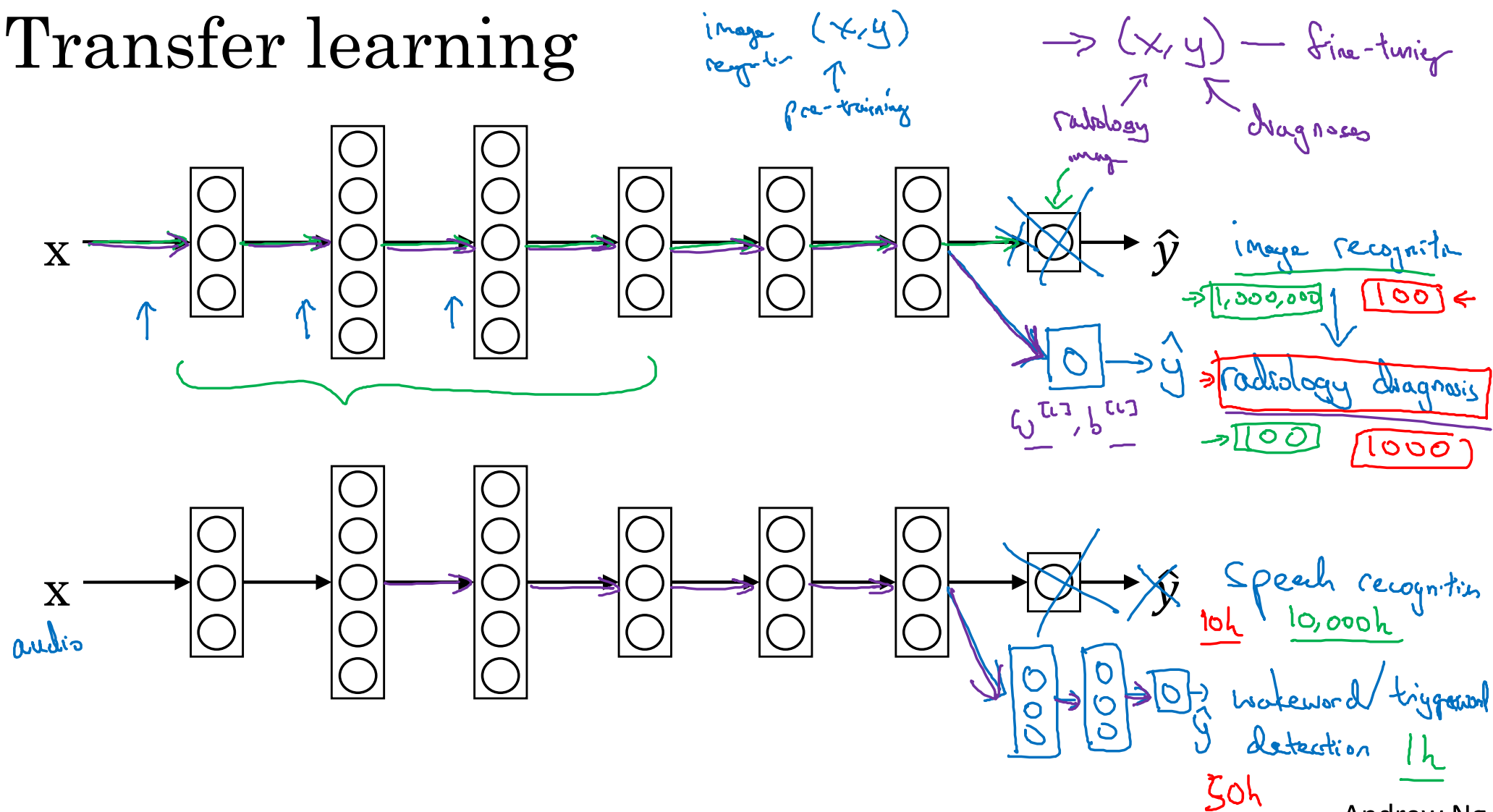


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Learning from
multiple tasks

Transfer learning

Transfer learning



When transfer learning makes sense

Transfer from A $\rightarrow$ B

- Task A and B have the same input x .
- You have a lot more data for Task A than Task B.

- Low level features from A could be helpful for learning B.



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Learning from
multiple tasks

Multi-task
learning

Simplified autonomous driving example



$x^{(i)}$

Pedestrians

Cars

Stop signs

Traffic lights

$\vdots$

$y^{(i)}$

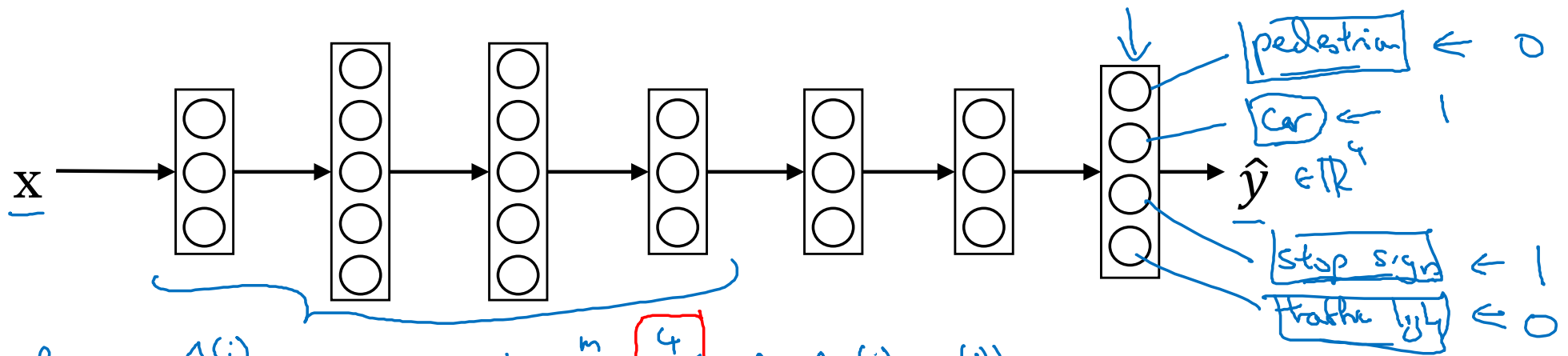
$(4, 1)$

$\begin{matrix} 0 \\ 1 \\ 1 \\ 0 \\ \vdots \end{matrix}$

$$Y = \begin{bmatrix} y^{(1)} & y^{(2)} & y^{(3)} & \dots & y^{(m)} \\ 1 & 1 & 1 & \dots & 1 \end{bmatrix}$$

$(4, m)$

Neural network architecture



Loss: $\mathcal{L}(\hat{y}^{(i)}, y^{(i)})$

$$\rightarrow \frac{1}{m} \sum_{i=1}^m \sum_{j=1}^4$$

Sum only over
value of j with
0/1 label.

Usual logistic loss

$$-y_j^{(i)} \log \hat{y}_j^{(i)} - (1 - y_j^{(i)}) \log (1 - \hat{y}_j^{(i)})$$

Unlike softmax regression:

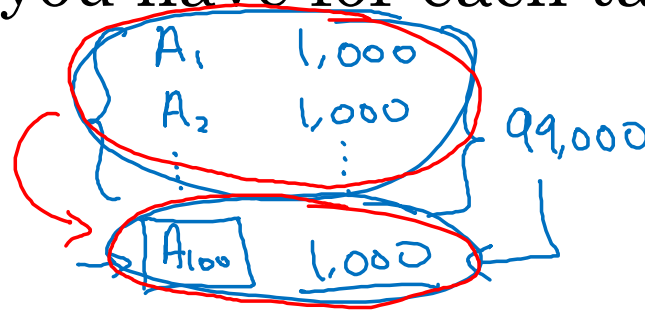
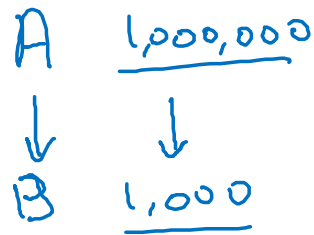
One image can have multiple labels

Multi-task learning

$$Y = \begin{bmatrix} 1 & 1 & 0 & ? \\ 0 & 1 & 1 & ? \\ ? & ? & 0 & ? \\ ? & ? & 0 & ? \end{bmatrix}$$

When multi-task learning makes sense

- Training on a set of tasks that could benefit from having shared lower-level features.
- Usually: Amount of data you have for each task is quite similar.



- Can train a big enough neural network to do well on all the tasks.



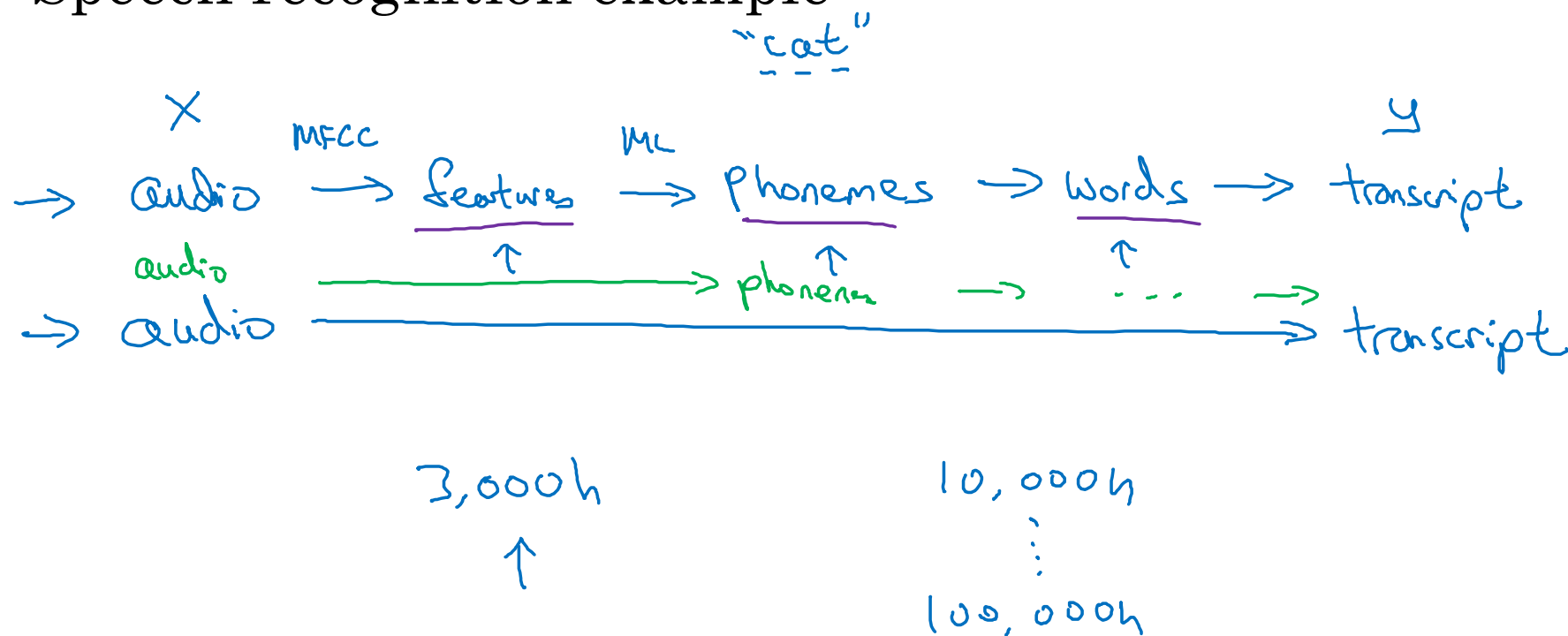
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End-to-end deep
learning

What is
end-to-end
deep learning

What is end-to-end learning?

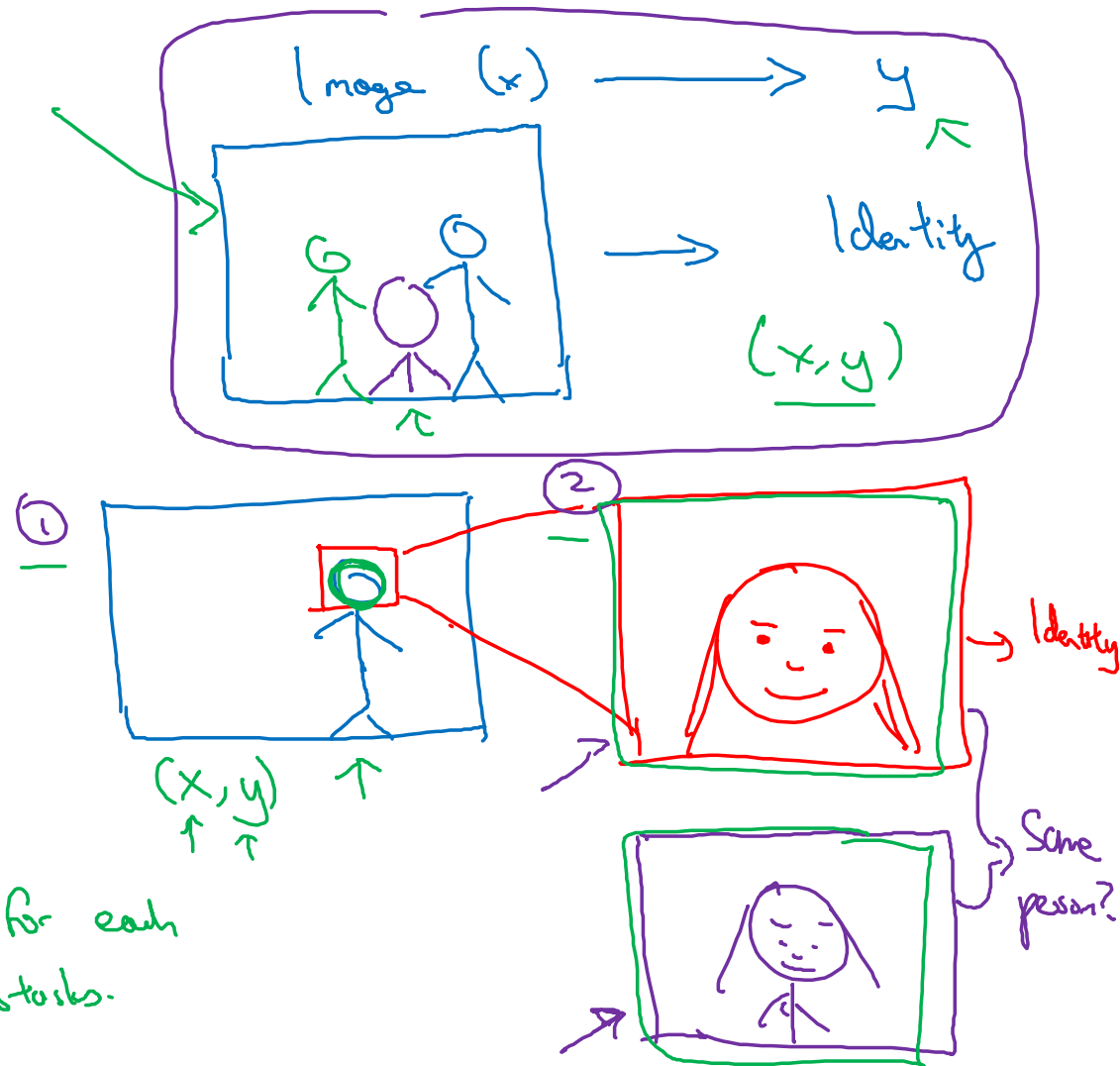
Speech recognition example



Face recognition



[Image courtesy of Baidu]



Have data for each
of 2 subtasks.

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More examples

Machine translation

(x, y)
English $\nearrow$ $\nwarrow$ French

English $\rightarrow$ text analysis $\rightarrow \dots \rightarrow$ French
English $\xrightarrow{\hspace{10em}}$ French

Estimating child's age:



Image $\xrightarrow{\textcircled{1}}$ bones $\xrightarrow{\textcircled{2}}$ age

Image $\xrightarrow{\hspace{10em}}$ age $\leftarrow$



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End-to-end deep
learning

Whether to use
end-to-end learning

Pros and cons of end-to-end deep learning

Pros:

- Let the data speak
- Less hand-designing of components needed

$x \rightarrow y$

→ "phonemes"
cat

Cons:

- May need large amount of data
- Excludes potentially useful hand-designed components

$x - - - - - \rightarrow y$

input
end
↓
 $x \rightarrow y$
output
end
↓

(x, y)

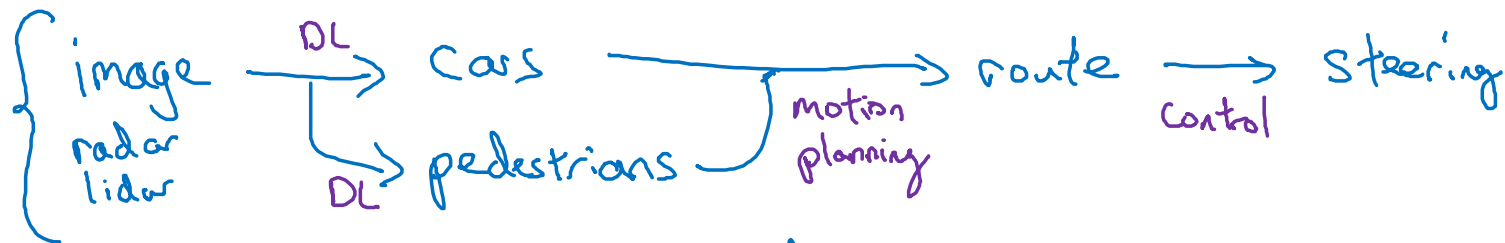
Data.
- - - -

Hand-design.
- - - -

Applying end-to-end deep learning

Key question: Do you have sufficient data to learn a function of the complexity needed to map x to y ?

$x \rightarrow y$



- Use DL to learn individual components
- Carefully choose $x \rightarrow y$ depending what tasks you can get data for.

$\rightarrow$ image $\longrightarrow$ steering